

POC Test Cases — Data Platform Validation



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Info

This document demonstrates the `testbook.cls` template for Huawei Cloud POC/acceptance test case documents. It exercises all available commands and environments: cover page, table of contents, test case environment with all field commands, objectives block, code blocks, images, callouts (`warning`, `tip`, `infobox`), tables, badge, menu paths, weblinks, notes, param references, and changelog.

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Test Cases Description

General Objective: Validate the Huawei Cloud Data Platform deployment for the POC environment, ensuring that all core services (DataArts Studio, MRS, OBS, VPC) are provisioned correctly, accessible, and functioning as specified in the solution architecture.

Prerequisites:

- Huawei Cloud account with IAM admin privileges in the `sa-brazil-1` region.
 - VPC, subnet, and security groups already created by the infrastructure team.
 - DataArts Studio instance (Dayu) provisioned and licensed.
 - MRS cluster deployed with the required component stack (HDFS, YARN, Spark, Hive).
 - OBS bucket created and accessible with the designated IAM policy.
 - Test execution environment with network connectivity to the Huawei Cloud Console.
 - `latexmk` and `terraform` CLI tools installed locally.
-

1.1 Test Scope

Domain	What Should Be Observed
Platform Architecture	Core services are deployed, healthy, and reachable via the Console API.
Data Engineering	DataArts Studio pipelines can ingest, transform, and load data into the target storage.
Data Storage	OBS buckets and MRS HDFS accept writes/reads with correct permissions.
Security	IAM policies, VPC isolation, and security group rules enforce the intended access control.

Table 1: Test scope by domain.

1.2 Preconditions and Preparations

- All infrastructure resources must be provisioned via Terraform and the `terraform apply` output must show zero errors.
- The MRS cluster must be in **Running** state before any data engineering test case is executed.
- Test data files (CSV/JSON) must be uploaded to the designated OBS bucket prior to pipeline execution.
- Each tester must have a personal IAM account with the **Dayu_User** role assigned in DataArts Studio.
- Network firewall rules must allow inbound TCP 22 (SSH) and TCP 443 (HTTPS) from the test execution subnet.



Ensure the test execution environment has the correct timezone set to America/Sao_Paulo (GMT-3) so that log timestamps align with the Huawei Cloud Console.

```
# Verify CLI tools are installed
terraform version
obsutil version
kubectl version --client
```

Tip

Use `terraform plan` before `terraform apply` to preview infrastructure changes and avoid unintended resource modifications.

1.3 Acceptance Method

Result	Explanation
Satisfied	The test case passes fully — observed behavior matches the expected result with no deviations.
Satisfied but with reservations	The test case passes but minor deviations were observed that do not impact the POC objectives (documented in Remarks).
Not satisfied	The test case fails — observed behavior does not match the expected result. A defect report must be filed.
Untested	The test case could not be executed due to blocking dependencies or environment issues.

Table 2: Acceptance criteria.



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Platform Architecture



2.1 TC-001: Verify MRS cluster deployment and health

Objective	Confirm that the MapReduce Service (MRS) cluster is deployed with the required components (HDFS, YARN, Spark, Hive) and all nodes report a healthy status.
Prerequisites	Terraform infrastructure apply completed successfully.
- MRS cluster ID is available in the Terraform output. - IAM user has MRS_Viewer or higher role.	
Procedure	Log in to the Huawei Cloud Console.
- Navigate to Console → MapReduce Service → Clusters. - Locate the cluster by its name (e.g. mrs-poc-cluster). - Verify the cluster status is Running. - Click the cluster name and review the Component tab — confirm HDFS, YARN, Spark, and Hive are listed with status Normal. - Review the Node tab — confirm all master and core nodes show status Running.	
	Page 6
	Cluster status is Running.



2.2 TC-002: Verify OBS bucket creation and accessibility

Objective	Confirm that the Object Storage Service (OBS) bucket used for the POC data lake is created, has the correct storage class, and is accessible with the designated IAM policy.
Prerequisites	1. OBS bucket name is defined in the Terraform configuration.
2. IAM policy granting read/write access to the bucket is attached to the test user.	
Procedure	1. Log in to the Console.
2. Navigate to Console → Object Storage Service.	
3. Locate the bucket (e.g. poc-datalake-raw).	
4. Verify the storage class is Standard.	
5. Upload a test file (test-upload.txt) to the bucket.	
6. Download the file and compare its content with the original.	
7. Delete the test file.	
Expected Result	1. Bucket exists with storage class Standard.
2. File upload completes without error.	
3. Downloaded file content	



2.3 TC-003: Verify VPC and security group configuration

Objective	Confirm that the VPC, subnet, and security groups are configured according to the network architecture diagram, with the correct CIDR blocks and inbound/outbound rules.
Prerequisites	1. VPC and subnet IDs are available from the Terraform output.
2. Security group names are documented in the solution architecture.	1. Log in to the Console.
Expected Result	1. VPC CIDR is 10.0.0.0/16.



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Data Engineering



3.1 TC-004: Verify DataArts Studio workspace creation

Required Objective	Confirm that the DataArts Studio (Dayu) instance is provisioned, the workspace is accessible, and the designated IAM users can log in with the Dayu_User role.
Prerequisites	1. DataArts Studio instance is provisioned and its status is Running.
2. Test user has the Dayu_User role assigned.	
Procedure	1. Log in to the Console with the test IAM user.
2. Navigate to Console → DataArts Studio → Workspaces.	
3. Locate the workspace (e.g. poc-workspace).	
4. Click Access Workspace to open the DataArts Studio console.	
5. Verify the left navigation panel loads with the expected modules: Data Integration, Data Development, Data Architecture.	
Expected Result	1. Workspace is listed and accessible.
2. DataArts Studio console opens without errors.	
3. All three modules (Data Integration, Data Development, Data	



3.2 TC-005: Verify data ingestion pipeline execution

Objective	Confirm that a DataArts Studio batch pipeline can successfully ingest CSV data from OBS, apply a basic transformation, and write the result to the target OBS path.
Prerequisites 2. Source CSV file (customers.csv) exists in the OBS raw bucket. 3. Target OBS path (poc-datalake-curated/customers/ is configured. 4. Pipeline pl-ingest-customers is published in DataArts Studio.	1. TC-004 completed (DataArts Studio workspace accessible).
Procedure 2. Locate the pipeline pl-ingest-customers in the job list. 3. Click Run to execute the pipeline. 4. Wait for the pipeline status to change to Success (timeout: 10 minutes). 5. Navigate to OBS and verify the output file exists in poc-datalake-curated/customers/ 6. Download the output file and	1. In the DataArts Studio console, navigate to Data Development.



3.3 TC-006: Verify Spark SQL query on MRS cluster

Objective	Confirm that a Spark SQL query can be executed on the MRS cluster to read data from a Hive table and return correct results, validating the integration between MRS and the data lake.
Prerequisites 2. Hive table <code>poc_db.customers</code> exists and contains data loaded by the ingestion pipeline. 3. SSH access to the MRS master node is configured.	1. TC-001 completed (MRS cluster healthy).
Procedure 2. Run the Spark SQL CLI: <code>spark-sql --master yarn --conf spark.sql.hive.convertMetastoreOrc=true -e "SELECT COUNT(*) FROM poc_db.customers;"</code> 3. Verify the returned count matches the expected row count (10,000). 4. Run a sample query to verify data integrity: <code>spark-sql -e "SELECT customer_id, name FROM poc_db.customers LIMIT 5;"</code> 5. Confirm the query returns 5 rows with non-null values.	1. SSH into the MRS master node.



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Security and Access Control



4.1 TC-007: Verify IAM policy enforcement on OBS buckets

New Objective	Confirm that IAM policies correctly restrict OBS bucket access — authorized users can read/write while unauthorized users are denied, and the bucket policy matches the principle of least privilege.
Prerequisites 2. Two test IAM users: one with OBS_ReadWrite role, one with no OBS permissions. 3. <code>obsutil</code> CLI configured for both users.	1. OBS bucket <code>poc-dataLake-raw</code> exists with the designated IAM policy.
Procedure 2. Using the authorized user, download and verify the file content. 3. Using the unauthorized user, attempt to upload a file — verify the request is denied (HTTP 403). 4. Using the unauthorized user, attempt to list objects in the bucket — verify the request is denied. 5. Review the bucket policy in the Console at Console → Object Storage Service → Bucket Policies and confirm it grants only	1. Using the authorized user, upload a test file to <code>poc-dataLake-raw</code> .



4.2 TC-008: Verify DataArts Studio data masking rule

Objective	Confirm that sensitive data columns (e.g. PII fields) are masked according to the DataArts Studio data masking rule when queried by non-admin users.
Prerequisites	1. TC-004 completed (DataArts Studio workspace accessible). 2. Data masking rule configured for column <code>customers.ssn</code> (Social Security Number). 3. Test user has <code>Dayu_User</code> role (not admin).
Procedure	1. In the DataArts Studio console, navigate to Data Architecture → Data Masking. 2. Verify the masking rule for <code>customers.ssn</code> is active and uses the Mask All strategy. 3. Query the <code>customers</code> table as the test user via DataArts Studio SQL console. 4. Verify the <code>ssn</code> column returns masked values (e.g. <code>*****</code>) instead of real data. 5. Query the same table as an admin user and verify real values are returned.
Expected Result	1. Masking rule is active for <code>customers.ssn</code>. 2. Non-admin user sees masked



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Environment Setup and Tooling



5.1 TC-009: Verify Terraform infrastructure deployment

Objective	Confirm that the Terraform configuration deploys all required infrastructure resources (VPC, MRS, OBS, DataArts Studio) without errors and the state file reflects the expected resources.
Prerequisites	1. terraform CLI installed (version $\geq$ 1.0).
	2. AK/SK credentials configured via environment variables or provider.tf. 3. Terraform configuration files in the project directory.
Procedure	1. Navigate to the Terraform project directory.
	2. Run <code>terraform init</code> to initialize the working directory. 3. Run <code>terraform plan</code> and review the planned changes — confirm all expected resources are listed. 4. Run <code>terraform apply -auto-approve</code> and wait for completion. 5. Run <code>terraform output</code> and verify all output values are populated. 6. Run <code>terraform state list</code> and confirm the resource count matches



5.2 TC-010: Verify test data upload to OBS

Objective	Confirm that all required test data files (CSV, JSON) can be uploaded to the designated OBS bucket and are accessible by the DataArts Studio pipeline.
Prerequisites 2. Test data files prepared in the local directory <code>./test-data/</code> . 3. <code>obsutil</code> CLI configured with the test user credentials.	1. TC-002 completed (OBS bucket accessible).
Procedure 2. Upload all files to the OBS raw bucket using <code>obsutil</code> . 3. Verify each file exists in OBS by listing the bucket contents. 4. Download a sample file from OBS and compare it with the local original (checksum). 5. In DataArts Studio, verify the data source connection can read the uploaded files.	1. List the local test data files and verify their integrity (row count, schema).
<pre># Upload test data files to OBS obsutil cp ./test-data/c</pre>	1. All test data files are uploaded without errors.

[Image placeholder]

File: assets/test-execution-dashboard.png

Description: Screenshot of the test execution dashboard showing all test case results

Place the image file at the path above, then replace this command with

*`\image{assets/test-execution-dashboard.png}` or
`\imagecap{assets/test-execution-dashboard.png}{caption}`.*

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Changelog

1.1.0

September 12, 2026

- Added comprehensive feature demonstrations: `warning`, `tip`, `infobox` callouts; `code blocks`; `badge`; `weblink`; `note`; `param`; `image`, `imagecap`, and `imageplaceholder`; `menu paths`; and `inlinecode` references.
- Added Security and Access Control test cases (TC-007, TC-008).
- Added Environment Setup and Tooling test cases (TC-009, TC-010).

1.0.0

September 12, 2026

- Initial version.
 - Added Platform Architecture test cases (TC-001 to TC-003).
 - Added Data Engineering test cases (TC-004 to TC-006).
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